



Proactive bill prediction and budget planning

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Introduction

Many households struggle to monitor their electricity consumption, often exceeding their budget before receiving the monthly bill. This project uses AI to predict electricity bills based on consumption data, the user's budget, and factors such as weather, while providing early alerts and personalised recommendations to reduce energy usage.

Our solution is an AI-powered Personal FinTech application designed to automate, govern, and optimize household operational expenditures, specifically targeting energy bills as the primary recurring financial burden. Built upon the ITU-T Y.3172 architectural framework for machine learning in future networks and strictly aligned with SDAIA's National AI Ethics Principles, the platform enables families to establish dynamic, climate-aware monthly consumption budgets.

By integrating real-time smart meter ingestion, localized weather telemetry, and predictive time-series modeling within a secure ML Sandbox, the system delivers precise, explainable bill forecasting and early budget threshold alerts. This proactive approach empowers households to manage cash flows effectively, prevent end-of-month financial shocks during peak summer cooling seasons, and maintain full transparency and privacy over their consumption data.

Technical Architecture & ML pipeline

Smart Financial Engine: Real-time algorithms converting raw kWh telemetry into SAR expenditure, factoring in official tariffs, VAT, and fixed fees.

Cloud Data Pipeline: Secure cloud backend that receives, cleanses, and streams real-time meter telemetry directly to the AI model.

LLM + RAG Assistant: An AI chatbot powered by Retrieval-Augmented Generation (RAG) over official energy regulations to deliver verified financial and saving advice.

Technical Architecture & ML Pipeline

The platform adopts the standard 7-node pipeline specified by ITU-T Y.3172:

Source Nodes (SRC): Ingests raw inputs (smart meter data, weather telemetry via APIs, and user budget caps).

Collector Node [C]: Aggregates incoming data streams in real time using lightweight protocols (MQTT and SBA Interfaces).

Preprocessor Node (PP): Cleanses data (smart gap-filling), enforces user privacy via anonymization, and extracts consumption behavior patterns.

Model Node (M): Runs time-series ML models to forecast total end-of-month energy expenses.

Policy Node (P): Compares real-time forecasts against budget caps to trigger automated alert logic.

Distributor Node (D): Delivers output actions and notifications to their target channels (e.g., push notifications).

Sink Node (SINK): The end-user mobile app interface that displays predictions, dashboards, and financial insights.

ML Sandbox & Lifecycle Orchestration (MLFO):

Isolated ML Sandbox: An isolated testing environment where models are calibrated and tested against severe weather spikes without triggering false user alerts.

Personalized Household Models: Models fine-tuned per household to reflect unique seasonal usage profiles.

ML Function Orchestrator (MLFO): Dynamically switches between models (e.g., summer vs. winter profiles) based on incoming weather telemetry.

Graduated Alerting Policy:

Level 1 (Threshold Alert): Sent upon reaching **50%** of the monthly budget.

Level 2 (Proactive Warning): Predictive alert dispatched if weather forecasts signal an impending budget breach.

Evaluation scenario

Step 1 (Deployment): The household sets a monthly energy budget of 500 SAR. The system ingests meter and weather telemetry seamlessly.

Step 2 (Event): A severe summer heatwave triggers a spike in cooling energy consumption, while an external party attempts unauthorized data access.

Step 3 (Enforcement & Mitigation): MLFO switches to the summer profile inside the ML Sandbox. The Policy node dispatches a Level 2 proactive warning, and PP anonymizes data to comply with PDPL regulations.

Policy gaps:

- No Unified Smart Meter API: Lack of official open APIs for third-party AI apps to access meter data directly without relying on IoT hardware.

- **Undefined Liability for Prediction Errors:** Current laws do not specify the legal liability of AI models if inaccurate bill forecasts lead to user financial miscalculations.
- **Need for Sector-Specific Energy Data Regulations:** While the PDPL provides a robust general framework, there is a lack of specific guidelines governing granular household consumption data.
- **No Algorithm Auditing Standards:** Lack of national frameworks to audit and certify AI energy models for accuracy, transparency, and fairness.
- **Absence of Dynamic Pricing Incentives:** Lack of regulations allowing flexible tariff structures, limiting the financial impact of AI savings recommendations.
- **Weak RAG Data Governance Frameworks:** No centralised, machine-readable repository for updated national energy regulations, risking outdated AI compliance.
- **No Standards for Carbon/Green Claims:** Absence of regulatory oversight to verify AI claims regarding household carbon footprint reduction.

Knowledge Base & Authentic Public References

KAPSARC Study on AI and Data Center Energy Efficiency (PDF):

<https://www.kapsarc.org/media/ip0dt4i1/dp00503-v1.pdf>

KAUST & Saudi Electricity Company (SEC) Joint Project for Smart Power Loss Reduction:

<https://www.kaust.edu.sa/news/kaust-helps-slash-sec-profit-losses-using-ml>

KBEAT Tool for Building Energy Analysis and Assessment in Saudi Arabia:

<https://apps.kapsarc.org/appboard/kbeat/en>

Research Paper on Machine Learning Algorithms for Energy Consumption Forecasting in Saudi Arabia:

<https://www.mdpi.com/1996-1073/16/4/2035>

Google DeepMind Case Study on AI-Driven Cooling Energy Optimization:

<https://deepmind.google/discover/blog/deepmind-ai-reduces-google-data-centre-cooling-energy-by-40/>

SDAIA National AI Ethics Principles:

<https://sdaia.gov.sa/>

Saudi Personal Data Protection Law (PDPL):

<https://www.google.com/search?q=https://sdaia.gov.sa/ar/SDAIA/about/Pages/AboutPDPL.aspx>

ITU-T Y.3172 Architectural Framework Standard:

<https://www.itu.int/rec/T-REC-Y.3172-201906-I>